✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 2

Radio Propagation Theory

Radio Navigation & Aids — Ground Training Series

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1. Introduction

> **What this section covers:** The definition of propagation, why different frequency bands use different paths, and how propagation paths determine both the uses and limitations of specific frequency bands.

In the context of radio waves, **propagation** simply means how radio waves travel through the atmosphere. Different frequency bands use different propagation paths; the propagation path often determines the uses to which a particular frequency band can be put in either communication or navigation systems, and also imposes limitations on the use of those frequencies.

2. Factors Affecting Propagation

> **What this section covers:** Six factors that affect how far and how reliably a radio wave reaches a receiver: attenuation (absorption + inverse square law), static interference, fading, transmitter power, receiver sensitivity, and directivity.

2.1 Attenuation

Attenuation is the term for the loss of signal strength in a radio wave as it travels outward from the transmitter. There are two distinct aspects:

Absorption

As a radio wave travels outward, its energy is **absorbed and scattered** by: molecules of air and water vapour, dust particles, water droplets, vegetation, the surface of the earth, and the ionosphere.

> ⚠ **Absorption Rule**
>
> The effect of absorption (except ionospheric absorption) **increases as frequency increases** and becomes a very significant factor above approximately **1000 MHz**.

Inverse Square Law

EM radiation from an aerial spreads out as the surface of a sphere, so power decreases with increasing distance.

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Power available is proportional to the inverse of the square of the range: $P \propto 1/R^2$

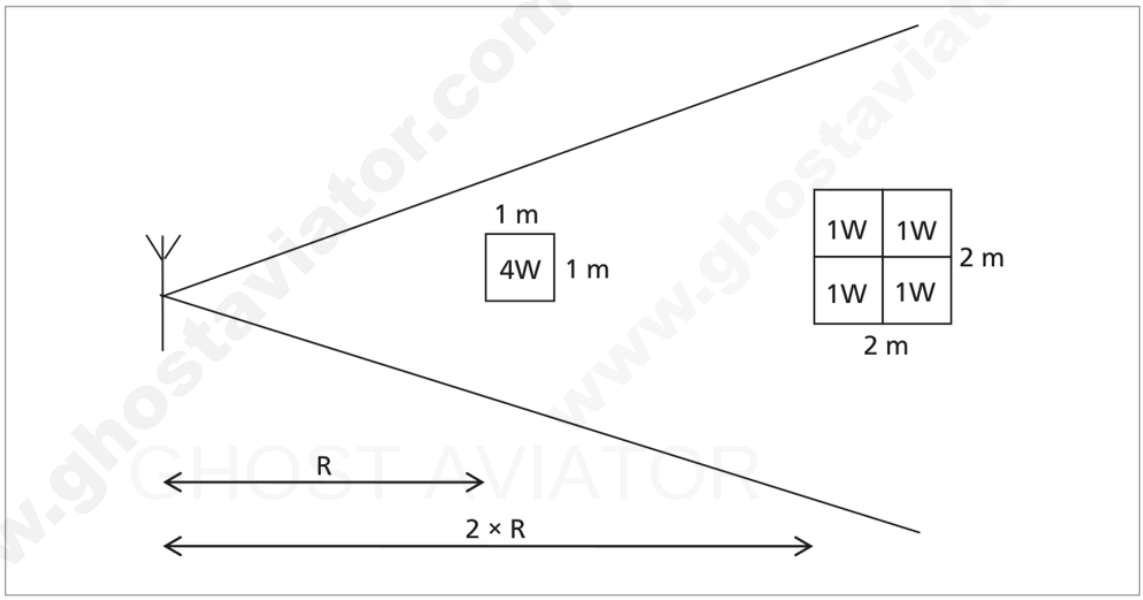


Figure 2.1 Inverse Square Law

Figure 2.1 — Inverse Square Law: at distance R, intensity = 4 W/m². At double the distance (2R), the same energy is spread over 4× the area, so intensity = 1 W/m²

Key Exam Implication

To **double the range** of a transmitter, the power output must be **increased by a factor of 4** ($2^2 = 4$). To triple the range → power × 9. To increase range by factor n → power × n^2.

Worked Example — Power & Range

`Q: Increase VHF range from 50 NM to 100 NM. Factor = 100/50 = 2. Power factor = 2² =` **4**

`Q: Increase range from 50 NM to 150 NM. Factor = 3. Power factor = 3² =` **9**

2.2 Static Interference

Large amounts of static electricity are generated in the atmosphere by weather, human activity, and geological activity. Key points:

- Effect of static interference is **greater at lower frequencies**

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- At **VHF and above**, interference is generally negligible
- Radio waves travelling through the **ionosphere collect interference at all frequencies**
- Receiver and transmitter circuitry also produce static interference

The strength of the required signal relative to interference is expressed as the **Signal-to-Noise Ratio (S/N)**. Best clarity/accuracy requires minimising the noise floor.

2.3 Fading

Transmissions following different paths (e.g. reflections) can arrive at a receiver simultaneously but not necessarily in phase. In extreme cases two signals arrive in **anti-phase** and cancel each other out. Signals going in and out of phase produce alternate fading and strengthening of the received signal.

2.4 Power

An increase in transmitter power output will increase range, within the limits of the inverse square law. To double the range → power × 4.

2.5 Receiver Sensitivity

If internal noise in a receiver can be reduced, the receiver will process weaker signals — increasing effective range. However, reducing receiver noise is an expensive process.

2.6 Directivity

Concentrating power into a **narrow beam** increases range or reduces power required for a given range. However, the signal will only be usable **in the direction of the beam**.

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3. Propagation Paths Overview

> **What this section covers:** The four propagation paths used in aviation: surface wave, space wave, sky wave, and satellite comms. Which frequencies use which path.

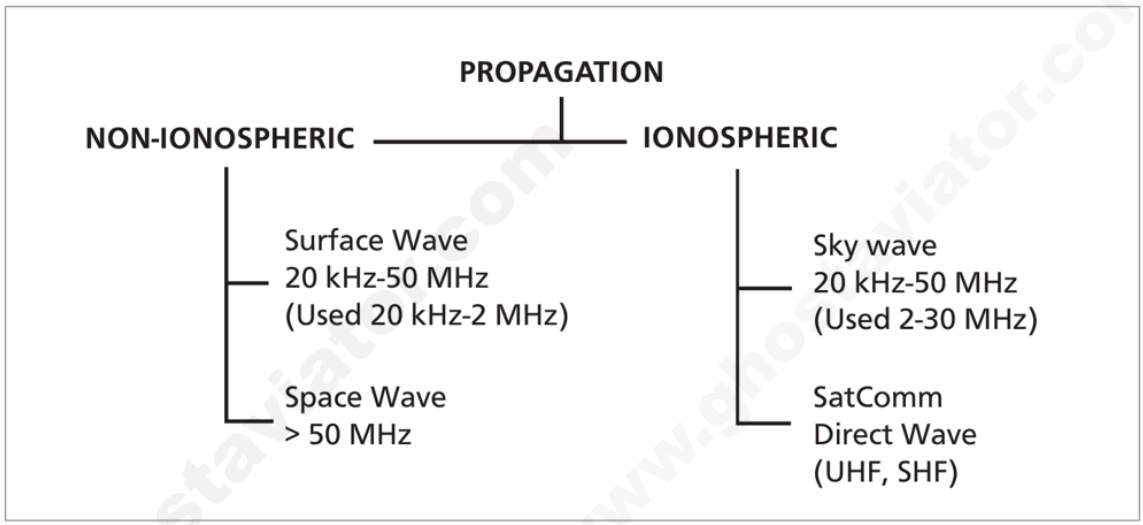


Figure 2.2

Figure 2.2 — Propagation paths: Non-ionospheric (Surface Wave 20 kHz–50 MHz; Space Wave >50 MHz) and Ionospheric (Sky Wave 20 kHz–50 MHz; SatComm/Direct Wave UHF, SHF)

PROPAGATION

NON-IONOSPHERIC

Surface Wave
20 kHz – 50 MHz
(Used: 20 kHz – 2 MHz)

Space Wave
> 50 MHz (VHF+)

IONOSPHERIC

Sky Wave
20 kHz – 50 MHz
(Used: 2 – 30 MHz / HF)

SatComm / Direct Wave
(UHF, SHF)

Ionospheric propagation is affected by the properties of the ionosphere. For this chapter the focus is on **sky wave**. Satellite propagation is covered in Chapter 18. Knowledge of propagation below **30 kHz** is not required.

4. Non-Ionospheric Propagation

4.1 Surface Wave

> **What this section covers:** How surface waves travel, diffraction, factors affecting surface wave range (frequency, terrain, polarization), and the range formula for MF transmissions.

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Surface wave propagation exists at frequencies from approximately **20 kHz to 50 MHz** (upper VLF to lower VHF). The portion of the wave in contact with the Earth's surface is retarded, causing the wave to **bend around the surface** — a process known as **diffraction**.

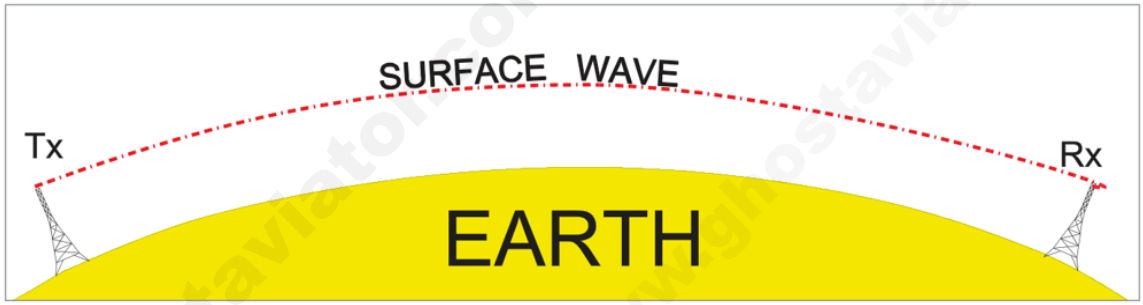


Figure 2.3 Surface Wave

Figure 2.3 — Surface Wave: the wave diffracts around the curvature of the Earth, following the surface from Tx to Rx

Factors Affecting Surface Wave Range

⚠ **Three Key Factors — All Examinable**

1. **Frequency:** As frequency increases → surface attenuation increases → surface wave range decreases. Surface wave is effectively non-existent above HF (**>30 MHz**).
2. **Surface type:** Losses are greater over land than over sea (sea has good electrical conductivity). Greater ranges are achievable **over sea**.
3. **Polarization:** A horizontally polarized wave is attenuated very quickly. Therefore **vertical polarization** is generally used at lower frequencies.

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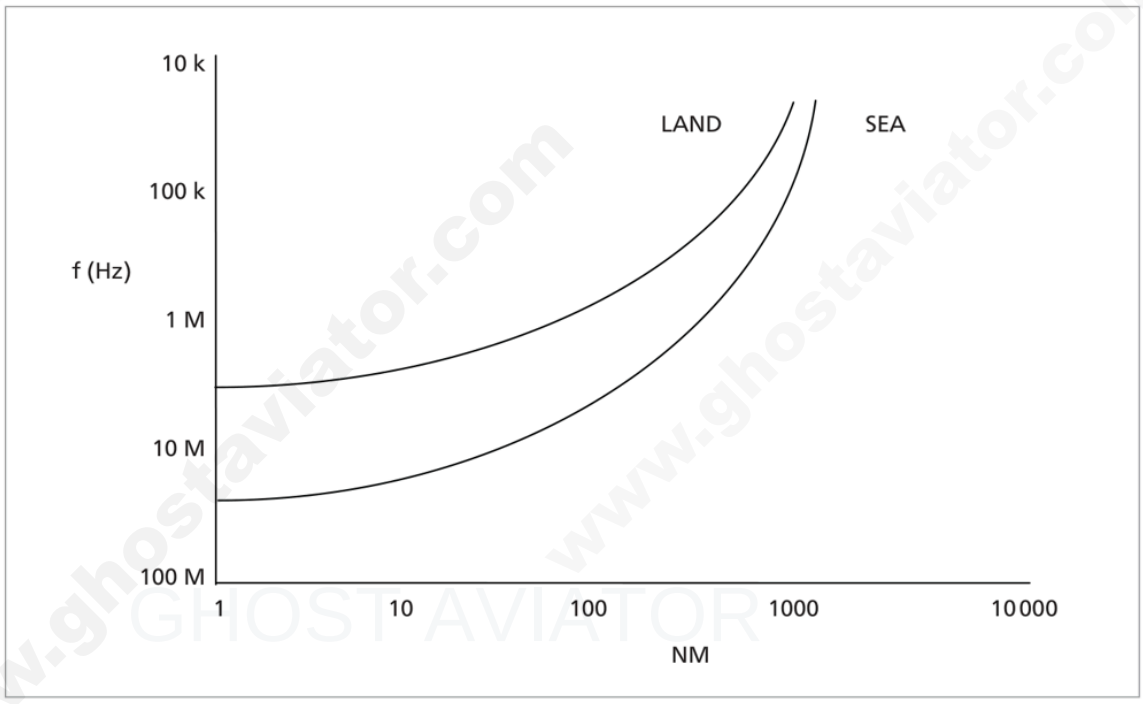


Figure 2.4

Figure 2.4 — Surface wave range (NM) vs frequency (Hz): sea propagation (upper curve) consistently outranges land propagation across all frequencies. Range decreases sharply as frequency rises above 1 MHz

This is the primary propagation path for the **LF frequency band** and the lower part of the **MF frequency band** (i.e. **30 kHz to 2 MHz**).

SURFACE WAVE RANGE FORMULA — MF AT 300 KHZ

Sea: Range ≈ 3 × √Power (kW) [NM]

Land: Range ≈ 2 × √Power (kW) [NM]

Worked Example

300 kHz transmitter, Power = 10 kW

Sea range = 3 × √10 = 3 × 3.162 = **~9.5 NM ×10 = ~300 NM**

Land range = 2 × √10 = 2 × 3.162 = **~6.3 ×10 = ~200 NM**

(√Power where Power in kW, result in hundreds of NM for 300 kHz)

4.2 Space Wave

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What this section covers: The two components of the space wave (direct + reflected), line-of-sight limitation at VHF and above, atmospheric refraction, and the LOS range formula.

The space wave comprises two paths: a **direct wave** and a **reflected wave**.

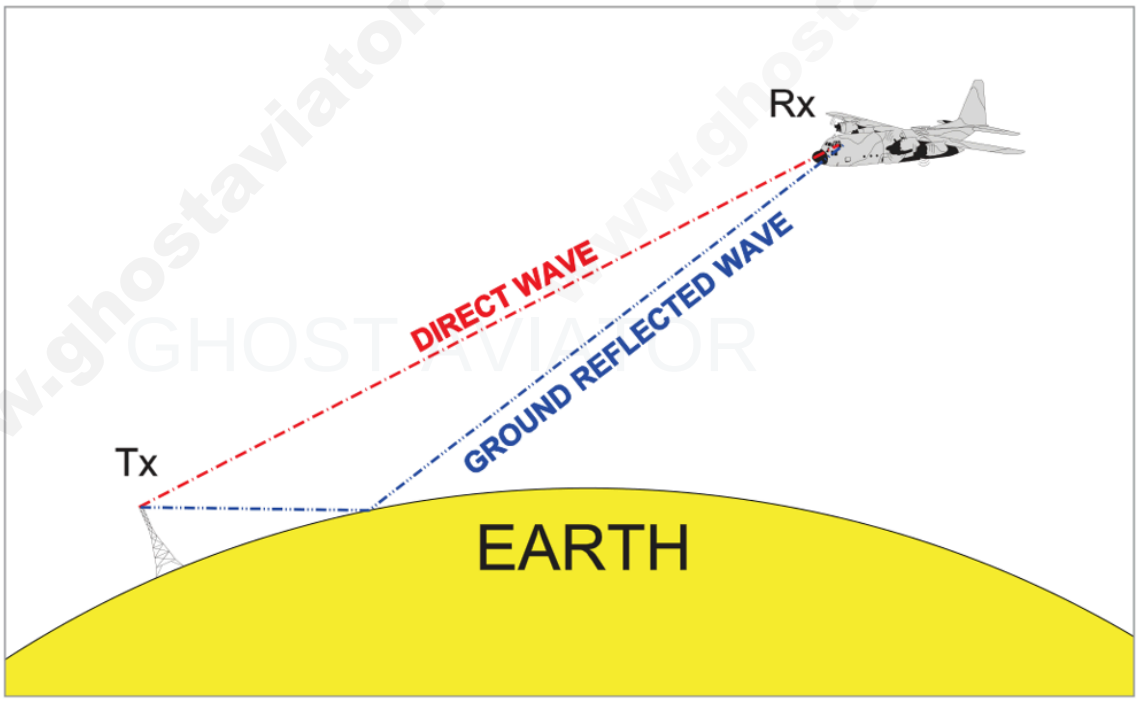


Figure 2.5 Space wave

Figure 2.5 — Space wave: the direct wave travels line-of-sight from Tx to Rx; the reflected wave bounces off the Earth's surface. Both arrive at the receiver simultaneously

At frequencies of **VHF and above**, radio waves start to behave more like visible light — there is a **radio horizon** just as there is a visual horizon. The only atmospheric propagation at these frequencies is **line of sight**.

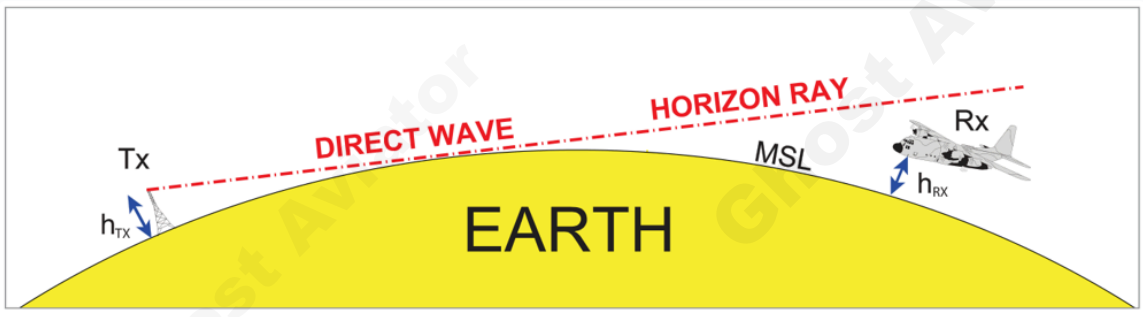


Figure 2.6 Maximum theoretical range

Figure 2.6 — Maximum theoretical LOS range: the direct wave path tangential to Earth, with Tx and Rx at defined heights. Atmospheric refraction extends range slightly beyond the geometric horizon

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> ⚠ **Critical VHF Rule**
>
> At VHF and above, regardless of transmitter power, if the receiver is **below the line-of-sight range**, it will receive **nothing**. Power is irrelevant below the radio horizon.

There is some **atmospheric refraction** which bends radio waves towards the Earth's surface, extending range slightly beyond the geometric horizon. For the EASA syllabus, the line-of-sight range formula is:

LOS RANGE FORMULA (VHF AND ABOVE)

$$\text{Range (NM)} = 1.23 \times (\sqrt{h_{TX}} + \sqrt{h_{RX}})$$

h_{TX} = transmitter height in **feet** | h_{RX} = receiver height in **feet**

> 📐 **Worked Examples**
>
> **Example 1:** Tx at 1024 ft, Rx at 1600 ft
>
> $\text{Range} = 1.23 \times (\sqrt{1024} + \sqrt{1600}) = 1.23 \times (32 + 40) = 1.23 \times 72 =$ **88.6 NM**
>
> **Example 2:** Find min height for aircraft to be detected at 200 NM by radar at 1700 ft
>
> $200 = 1.23 \times (\sqrt{1700} + \sqrt{h_{AC}})$
>
> $200/1.23 = 162.6 = \sqrt{1700} + \sqrt{h_{AC}} = 41.23 + \sqrt{h_{AC}}$
>
> $\sqrt{h_{AC}} = 162.6 - 41.23 = 121.37$
>
> $h_{AC} = 121.37^2 \approx$ **14,731 ft ≈ 15,000 ft**

> 💡 **Exam Tip — Note 1**
>
> "*Regardless of the possible propagation paths, if a receiver is in line of sight with a transmitter, then the space wave will be received.*" This means at lower frequencies where surface wave or sky wave exist, the space wave is ALSO received if LOS exists — the space wave is always present when LOS is available.

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5. The Ionosphere

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What this section covers: Formation of the ionosphere, why ionization occurs, what affects ionization intensity (time of day, season, latitude, solar flares), and the dangers of solar flare activity to aviation systems.

The ionosphere extends upward from approximately **60 km** to the limits of the atmosphere (notionally **1500 km**). At these altitudes, atmospheric pressure is very low (**0.22 hPa at 60 km**) and gaseous atoms are widely dispersed.

Incoming **solar radiation at ultra-violet and shorter wavelengths** interacts with atoms, raising their energy levels and causing **electrons to be ejected** from atomic shells. Since an atom is electrically neutral, the result is:

- **Negatively charged free electrons**
- **Positively charged ions** (the remaining part of the atom)

Ionization Intensity — Key Relationships

🔵 **Factors Controlling Ionization Levels**

- **Time of day:** Highest levels ~ **1400 local time** (balance of ionization vs decay); lowest just before sunrise
- **Season:** Higher in **summer** than winter (greater solar radiation intensity)
- **Latitude:** Increases as latitude **decreases** (closer to equator = more solar intensity)
- **Solar flares:** Unpredictable but can cause **exceptionally high** ionization levels

⚠️ **Solar Flare Hazard to Aviation**

Intense solar flare activity → severe disruption of communication and navigation systems, particularly **space-based systems**. It is not unusual for communication satellites to be shut down during periods of intense solar flare activity to avoid damage.

Ionization levels do **not** increase linearly with altitude. Gravitation and terrestrial magnetism affect the distribution of gases at high altitudes (where normal atmospheric mixing is absent), causing ionized particles to form into **discrete layers**.

6. Ionospheric Layers (D, E, F)

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What this section covers: The three ionospheric layers, their average altitudes, day/night behaviour, seasonal variations, and how ionization is not uniform within each layer.

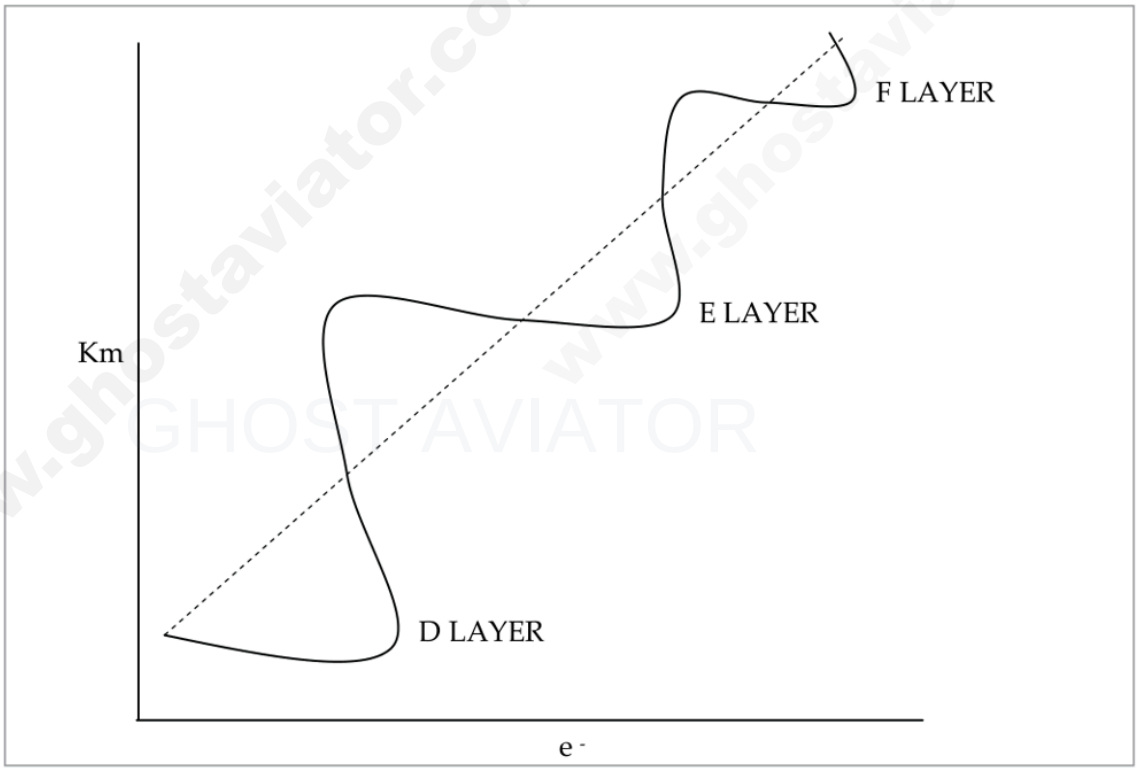


Figure 2.7 Effect of ionisation with height

Figure 2.7 — Effect of ionisation with height: D layer (lowest, ~75 km), E layer (~125 km), F layer (~225 km). Ionization is most intense at the centre of each layer

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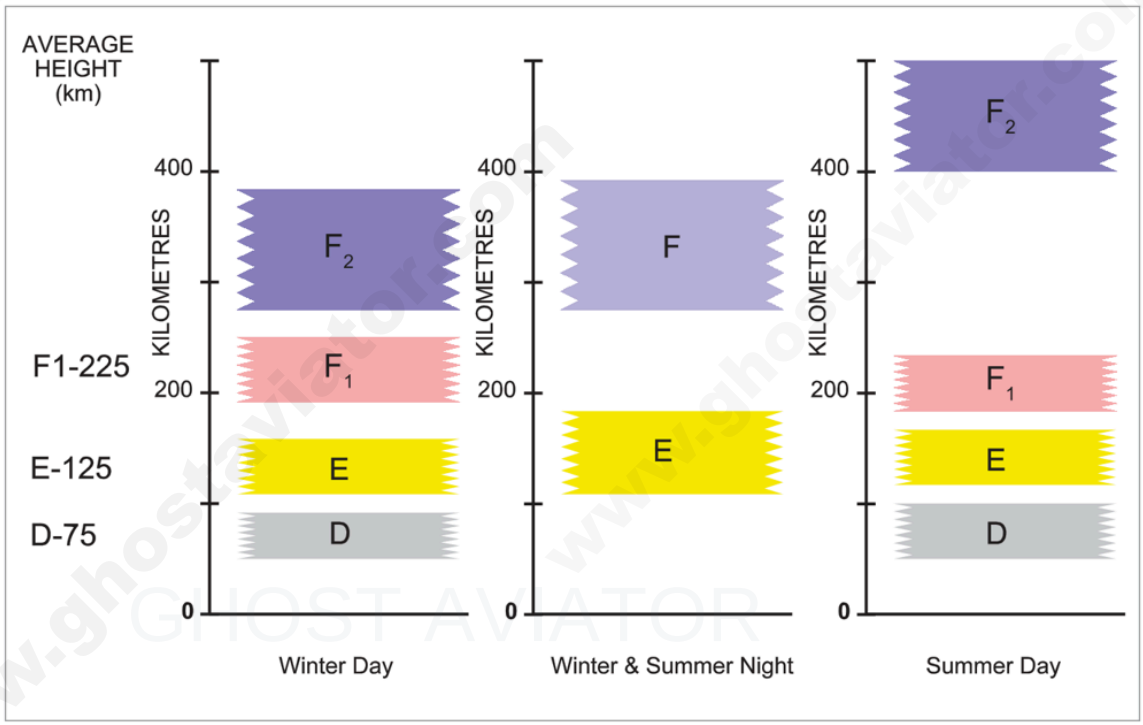


Figure 2.8 Layers of the ionosphere

Figure 2.8 — Ionospheric layer heights across Winter Day, Winter/Summer Night, and Summer Day. Note the F layer splitting into F1 and F2 in summer and the D layer disappearing at night

Layer	Avg Altitude	Day Behaviour	Night Behaviour	Aviation Significance
D Layer	~75 km	Forms at sunrise; absorbs LF/MF frequencies below ~2 MHz	Disappears at sunset	Blocks sky wave during day; night absence allows LF/MF sky wave
E Layer	~125 km	Reduces in altitude at sunrise	Increases in altitude after sunset; present 24 hours	Refracts sky wave up to ~2 MHz; max sky wave range ~ 1350 NM
F Layer	~225 km	Splits into F1 and F2 at sunrise	Rejoins at sunset	Refracts 2–50 MHz sky wave; max range ~ 2200 NM
F2 (summer)	Up to >400 km	Increases in altitude in summer	Reduces in winter	Greater skip distances in summer

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> ⚠ **Dawn & Dusk Instability**
>
> Around **dawn and dusk** the ionosphere is in a transitional state — "electrical turbulence." Radio navigation and communication systems using the ionosphere are subject to **excessive interference and disruption** during these periods.

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7. Sky Wave Propagation

7.1 Critical Angle & Skip Distance

> **What this section covers:** How ionization refracts radio waves back to Earth, the concept of the critical angle, skip distance, and what "first returning sky wave" means.

As a radio wave transits an ionized layer, it encounters an increasing density of ions toward the layer centre, then decreasing density toward the far edge:

- Waves at **right angles to the layer** → retarded but maintain a straight path (pass through)
- Waves at an **angle** → refracted away from the normal entering, refracted back toward normal exiting

If the radio wave refracts to the **Earth horizontal** before reaching the layer centre → it undergoes **total internal refraction** and returns to the Earth's surface as **sky wave**.

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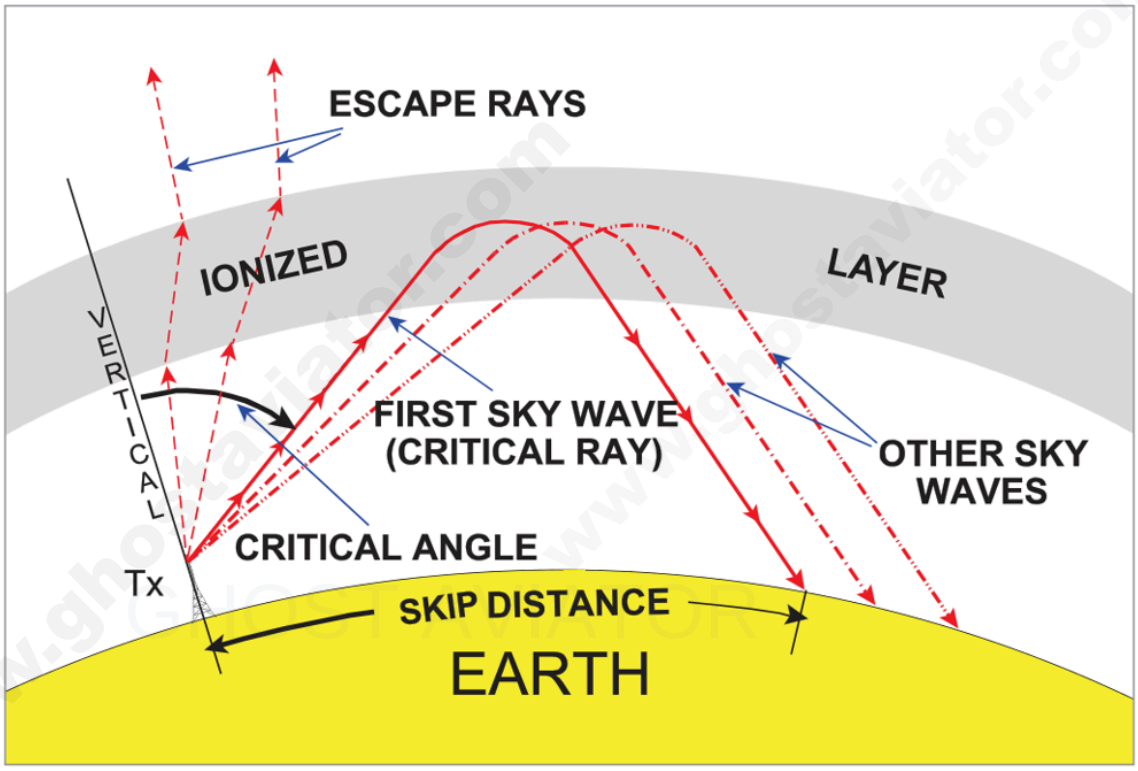


Figure 2.9 Sky wave propagation - critical angle

Figure 2.9 — Sky wave critical angle: waves at a shallow angle (escape rays) penetrate through; at the critical angle, total internal refraction returns the wave to Earth as the first returning sky wave. The skip distance is from Tx to the landing point of the first returning sky wave

🔵 **Key Definitions — Must Know**

- **Critical angle:** The angle from the vertical at which the first sky wave returns to Earth (total internal refraction occurs)
- **Skip distance:** The distance from the transmitter to the point where the first returning sky wave appears at the surface
- **Dead space:** The area between where the surface wave is totally attenuated and where the first returning sky wave appears — **no detectable signal**

7.2 Dead Space

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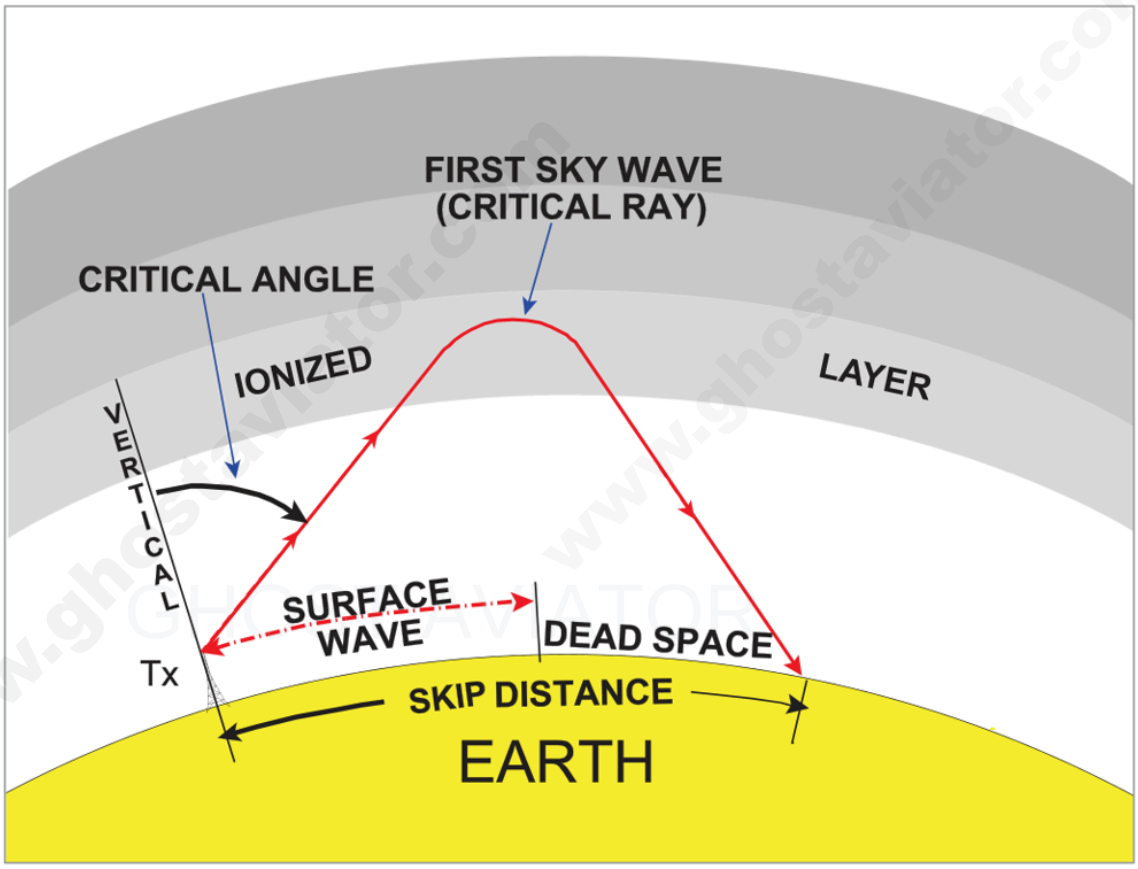


Figure 2.10 Sky wave propagation - dead space

Figure 2.10 — Dead space: the surface wave fades at a certain range; the first sky wave (critical ray) lands at the skip distance. Between these two points is the dead space — a zone of silence

Sky wave occurs in the **LF, MF, and HF** frequency bands. Frequencies above **50 MHz** (VHF) do **not** produce sky waves under normal conditions — VHF navigation systems are therefore free of sky wave interference.

> *Sky wave only likely above 50 MHz during abnormal ionospheric conditions associated with intense sunspot or solar flare activity.*

7.3 Effect of Changes in Ionization & Frequency

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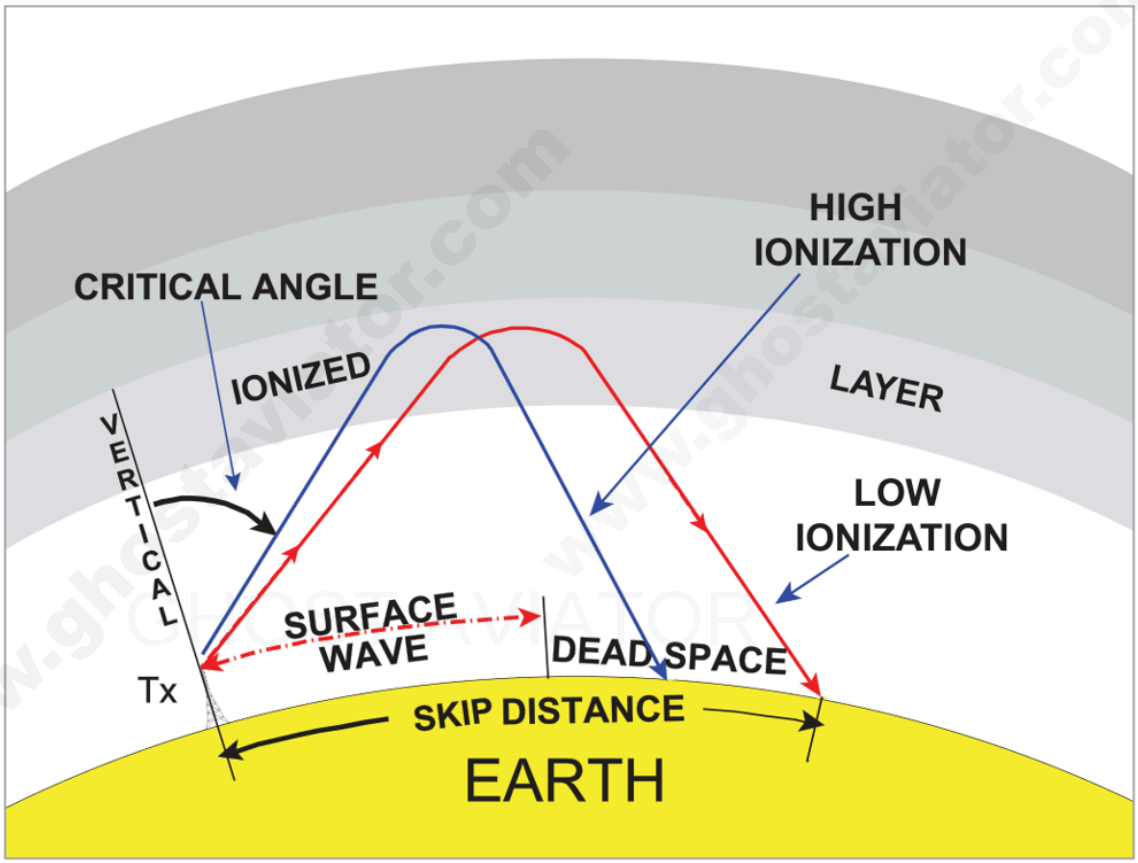


Figure 2.11 Sky wave propagation - effect of increased ionization

Figure 2.11 — High vs low ionization: higher ionization (solid line) returns the sky wave at a shorter skip distance; lower ionization (dashed) gives a longer skip distance and larger dead space

Change	Critical Angle	Skip Distance	Dead Space	Surface Wave Range
↑ Ionization	Decreases (smaller)	Decreases	Decreases	No direct effect
↓ Ionization	Increases	Increases	Increases	No direct effect
↑ Frequency	Increases	Increases	Increases	Decreases
↓ Frequency	Decreases	Decreases	Decreases	Increases
↑ Layer altitude	—	Increases	Increases	No direct effect

> 💡 **Memory Aid — Frequency & Dead Space**
>
> As frequency **increases**: skip distance ↑ AND surface wave range ↓. Both effects make the **dead space larger** from both ends simultaneously. This is the most tested relationship in this chapter.

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7.4 LF & MF Sky Wave

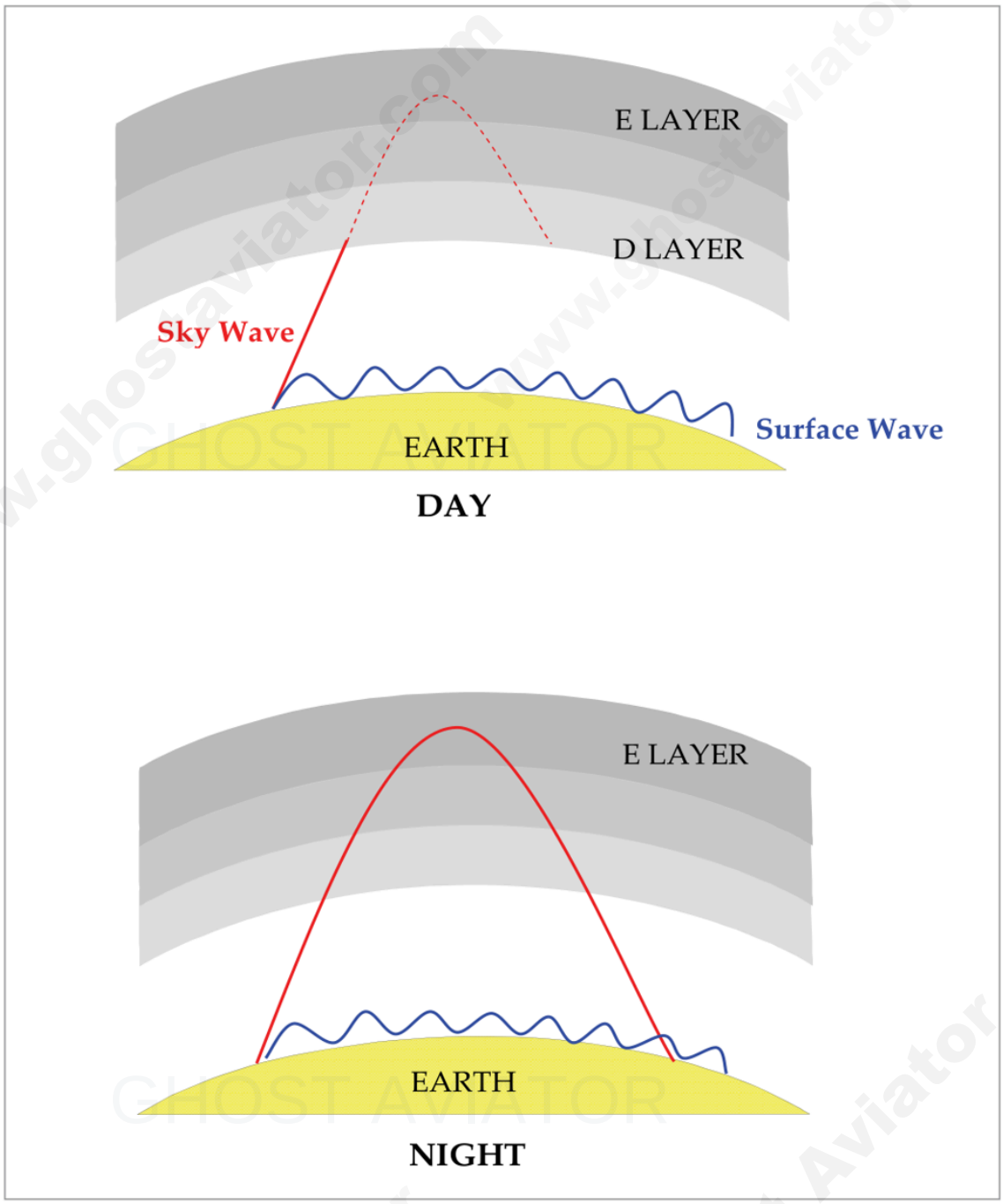


Figure 2.12 LF/MF Sky wave propagation

Figure 2.12 — LF/MF sky wave: DAY (top) — D layer absorbs sky wave below ~2 MHz, only surface wave reaches receiver; NIGHT (bottom) — D layer absent, E-layer sky wave present causing interference with surface wave navigation systems

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⚠ **Exam-Critical: LF/MF Night-time Sky Wave Problem**

During the day: D-layer absorbs sky wave at frequencies below ~ **2 MHz**. Only surface wave is present for LF/MF — reliable for NDB/ADF navigation.

At night: D-layer disappears. Sky waves refracted from the E-layer appear at relatively short ranges and cause **interference with short-range navigation systems** (NDB/ADF) relying on surface wave reception. This makes **ADF unreliable at night** — covered in detail in Chapter 7.

7.5 Achievable Ranges & Multi-hop

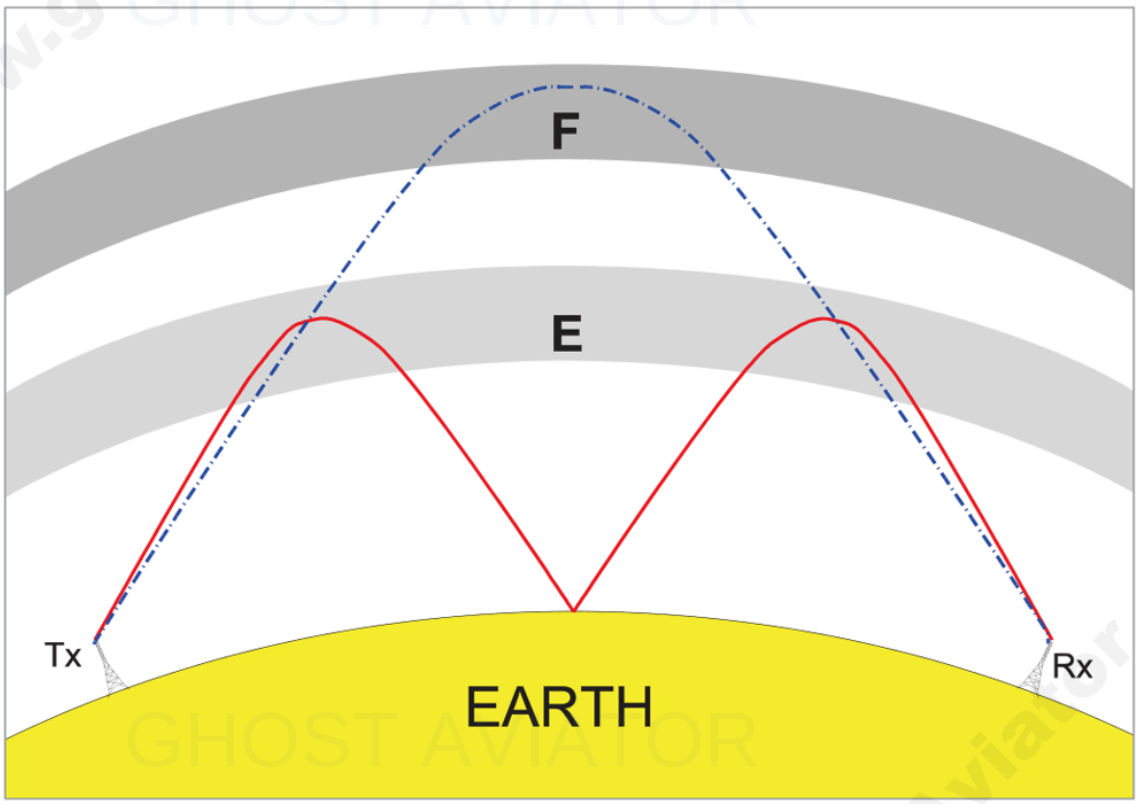


Figure 2.13 Multi-hop Sky wave propagation

Figure 2.13 — Multi-hop sky wave: the sky wave is refracted at the ionosphere, reflects off the Earth's surface, then refracted again at the ionosphere. Multiple hops can achieve global ranges

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✅ **Sky Wave Maximum Ranges**

- **E-layer** (avg **125 km**): average max sky wave range = **1350 NM**
- **F-layer** (avg **225 km**): average max sky wave range = **2200 NM**
- **Multi-hop sky wave:** wave refracted at ionosphere → reflected from Earth → refracted again. Can achieve ranges of **half the diameter of the Earth**

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8. HF Communications (MUF & OWF)

What this section covers: Why HF sky wave is currently the only practical long-range communication method over oceans, the Maximum Usable Frequency (MUF), the Optimum Working Frequency (OWF), day/night frequency selection, range vs frequency selection, and the classic mid-Atlantic HF example.

Over oceans and uninhabited land areas, VHF is impractical (line-of-sight limitation). To achieve ranges of **2000–3000 NM**:

- Surface wave at these ranges requires VLF/LF frequencies → complex, heavy equipment, high static, low data rates
- Currently the only practical solution is **HF Communications using sky wave propagation**

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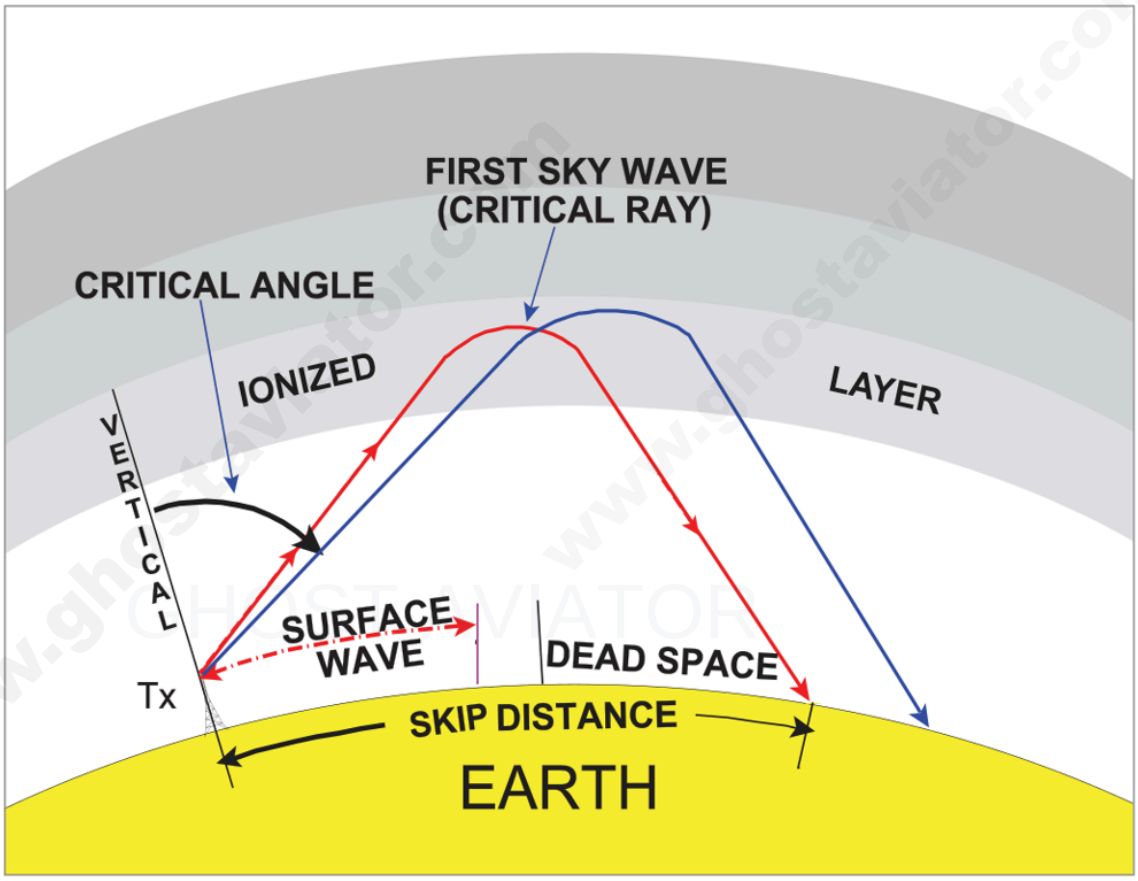


Figure 2.14

Figure 2.14 — HF sky wave comms: skip distance, surface wave extent, dead space, and the relationship between transmitter, ionospheric refraction point, and receiver

Maximum Usable Frequency (MUF)

The **MUF** for a given range is the frequency of the **first returning sky wave**. This is the ideal frequency because it has had the shortest path through the ionosphere — less attenuation, less static interference.

> ⚠ **Problem with Using MUF Directly**
>
> Since ionization levels fluctuate, a decrease in ionization increases the skip distance. If operating at the MUF, even a small reduction in ionization can cause loss of signal. Therefore MUF is not used as the operating frequency.

Optimum Working Frequency (OWF)

OWF = ***0.85 × MUF*** *— determined by decades of experimentation and experience.*

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The OWF provides a safety margin below the MUF so that normal ionization fluctuations do not cause loss of signal.

Day vs Night Frequency Selection

> 🔵 **Day/Night Rule of Thumb**
>
> Ionization is lower at night → frequency required at night is **roughly half** the frequency required by day for the same range.

Range vs Frequency Selection

> ✅ **Range–Frequency Relationship**
>
> - **Short ranges** → require **lower frequencies** (smaller skip distance needed)
> - **Long ranges** → require **higher frequencies** (larger skip distance needed)

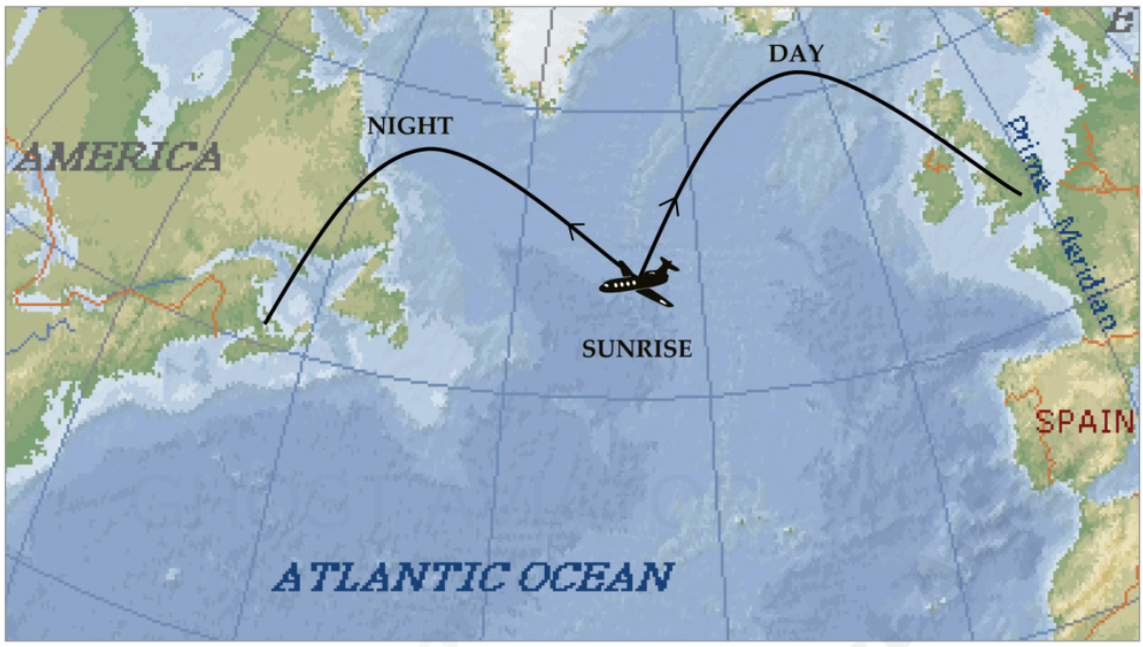


Figure 2.15 HF Communications Mid-Atlantic

Figure 2.15 — Mid-Atlantic HF scenario: Aircraft at sunrise, communicating with UK (day side) on 12 MHz and needing to select frequency for USA (night side). The refraction point midway to UK is in day; midway to USA is in night

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💡 Classic Exam Scenario — Mid-Atlantic at Sunrise

An aircraft in mid-Atlantic at sunrise communicates with UK on **12 MHz**.
What frequency to communicate with USA?

Answer: 6 MHz

Reasoning: Midway between aircraft and UK → it is **day** (high ionization → high frequency needed). Midway between aircraft and USA → it is **night** (low ionization → roughly half the frequency). Half of 12 = **6 MHz**.

9. Propagation Summary

What this section covers: A consolidated table of which propagation path is used (and which is present but not normally utilised) for each frequency band.

Frequency Band	Primary Path	Secondary Path (present but not normally used)
LF	Surface Wave	(Sky Wave)
MF	Surface Wave	(Sky Wave)
HF	Sky Wave	(Surface Wave)
VHF	Space Wave	—
UHF	Space Wave	—
SHF	Space Wave	—
EHF	Space Wave	—

10. Super-refraction & Sub-refraction

What this section covers: Abnormal atmospheric refraction at VHF and above — super-refraction (extended range, duct propagation) and sub-refraction (reduced range), their causes, and where they occur.

10.1 Super-refraction

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Significant at frequencies **above 30 MHz** (VHF and above). Radio waves experience **greater than normal downward bending** towards the Earth's surface, giving range increases of up to **40% above the usual LOS range**.

✅ **Conditions for Super-refraction**

- Decrease in relative humidity with height
- Temperature falling more slowly with height than standard (reduced lapse rate)
- Fine weather and **high pressure systems**
- Warm air flowing over a cooler surface

🔵 **Duct Propagation**

In extreme cases — a **low-level temperature inversion** with marked decrease in humidity with height (warm dry air above cool moist air) — a **low-level duct** forms that traps radio waves above 30 MHz, giving **extremely long ranges**. Classic real-world example: UK television interference from continental stations during summer high pressure.

Most common locations: warm desert areas bordering oceanic areas — **Mediterranean** and **Caribbean** seas. Also in temperate latitudes during winter high pressure (descending adiabatically warmed air over cool moist surface air).

10.2 Sub-refraction

Much rarer than super-refraction. Causes a **reduction** in normal refraction, giving a decrease in LOS range of up to **20%**.

⚠️ **Conditions for Sub-refraction**

- Increase in relative humidity with increasing height
- Temperature decreasing with height at a **greater rate than standard** (super-adiabatic lapse rate)
- Poor weather with **low pressure systems**
- Cold air flowing over a warm surface

Normal Refraction	Super-refraction	Sub-refraction
(Standard LOS Range)	Range +40%	Range −20%

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High pressure · Warm/dry over cool/moist · Duct possible

Low pressure · Cold over warm · High lapse rate

Feature	Super-refraction	Sub-refraction
Range effect	**Up to +40%**	**Up to −20%**
Frequency	>30 MHz (VHF+)	>30 MHz (VHF+)
Pressure	High pressure	Low pressure
Humidity profile	Decreases with height	Increases with height
Lapse rate	Less than standard	Greater than standard
Air movement	Warm over cool	Cold over warm
Extreme form	Duct propagation	N/A

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⚡ QUICK REVISION SUMMARY — CHAPTER 2

- **Attenuation:** absorption + inverse square law ($P \propto 1/R^2$; double range → power ×4)
- **Absorption** increases with frequency; significant above **1000 MHz**
- **Static** worse at lower frequencies; negligible at VHF+ (unless through ionosphere)
- **Surface wave:** 20 kHz–50 MHz; vertical polarisation; sea > land range; used 30 kHz–2 MHz
- **Space wave (VHF+):** line-of-sight only; Range = $1.23 \times (\sqrt{h_{TX}} + \sqrt{h_{RX}})$; power irrelevant below horizon
- **Ionosphere:** 60–1500 km; D (~75 km), E (~125 km), F (~225 km)
- **D layer:** day only; absorbs <2 MHz; absence at night → LF/MF sky wave interference
- **Critical angle** → skip distance → dead space (zone of silence)
- ↑ Ionization → skip distance ↓; ↑ Frequency → skip distance ↑, surface wave range ↓, dead space ↑↑
- **E-layer max range: 1350 NM; F-layer max range: 2200 NM**
- **OWF = 0.85 × MUF**; night frequency ≈ ½ day frequency; short range → lower frequency
- **Super-refraction:** +40% range; high pressure; duct possible; Mediterranean/Caribbean
- **Sub-refraction:** −20% range; low pressure; cold over warm

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📝 Practice Questions & Detailed Answers

How to Use This Section

Cover the answer panel and attempt each question independently. Read the explanation and distractor analysis thoroughly — this is where understanding is built, not just answers confirmed.

Q1. The process which causes the reduction in signal strength as range from a transmitter increases is known as:

a. absorption

b. diffraction

c. attenuation

d. ionisation

✓ Correct Answer: **(c)** — attenuation

Explanation: Attenuation is the overall term for loss of signal strength as range increases. It has two components: absorption (energy absorbed by matter) and the inverse square law (energy spread over increasing area). See Section 2.1.

Why the other options are wrong:

- **(a)** — Absorption is one component of attenuation, not the overall term. The question asks for the term for signal strength reduction with range — that is attenuation.
- **(b)** — Diffraction is the bending of radio waves around the Earth's surface (surface wave). It does not cause signal strength reduction with range.
- **(d)** — Ionisation is the process of creating ions in the upper atmosphere. It is a cause of some signal modification but is not the term for signal

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strength reduction with range.

Instructor's Note: *Classic definition trap. Absorption = one mechanism of attenuation. Attenuation = the overall effect (includes both absorption and inverse square law). The question asks for the overall term.*

Q2. Which of the following will give the greatest surface wave range?

a. 243 MHz

b. 500 kHz

c. 2182 kHz

d. 15 MHz

✓ Correct Answer: **(b)** — 500 kHz

Explanation: Surface wave range **decreases as frequency increases**. The lowest frequency in the list gives the greatest surface wave range. 500 kHz (MF) < 2182 kHz (MF) < 15 MHz (HF) < 243 MHz (VHF — no surface wave). See Section 4.1.

Why the other options are wrong:

- **(a)** — 243 MHz is VHF; surface wave is effectively non-existent above HF (30 MHz). This gives virtually zero surface wave range.
- **(c)** — 2182 kHz is the MF distress frequency (2.182 MHz). Higher than 500 kHz → shorter surface wave range.
- **(d)** — 15 MHz is HF; surface wave exists but is greatly attenuated — far less range than MF or LF.

Instructor's Note: *Surface wave is maximised at the lowest frequency. Always pick the lowest frequency for maximum surface wave range. The rule: lower frequency = less surface attenuation = more range.*

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Q3. It is intended to increase the range of a VHF transmitter from 50 NM to 100 NM. This will be achieved by increasing the power output by a factor of:

a. 2

b. 8

c. 16

d. 4

✓ Correct Answer: **(d)** — 4

Explanation: By the inverse square law, power ∝ $1/R^2$. To double the range (100/50 = factor of 2), power must increase by 2^2 = **4**. See Section 2.1.

Why the other options are wrong:

- **(a)** — A power factor of 2 would only increase range by √2 ≈ 1.41, i.e. from 50 to ~70 NM, not 100 NM.
- **(b)** — A factor of 8 would increase range by √8 ≈ 2.83, giving ~141 NM — more than required.
- **(c)** — A factor of 16 would increase range by √16 = 4, giving 200 NM — far more than required.

Instructor's Note: *Formula to remember: Power factor = (new range / old range)². Here: $(100/50)^2 = 2^2 = 4$. This is a direct application of the inverse square law — perhaps the most common calculation in this chapter.*

Q4. The maximum range an aircraft at 2500 ft can communicate with a VHF station at 196 ft is:

a. 79 NM

b. 64 NM

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c. 52 NM

d. 51 NM

✓ Correct Answer: **(a)** — 79 NM

Explanation: Range = 1.23 × (√h_{TX} + √h_{RX}) = 1.23 × (√196 + √2500) = 1.23 × (14 + 50) = 1.23 × 64 = **78.72 ≈ 79 NM**. See Section 4.2.

Why the other options are wrong:

- **(b)** — 64 NM = 1.23 × 52 — this is missing one of the square root values, or confusing the sum (64) with the range.
- **(c)** — 52 NM ≈ 1.23 × 42.3 — doesn't correspond to the correct square roots.
- **(d)** — 51 NM is close to 1.23 × 41.5 — possibly a calculation using wrong heights.

Instructor's Note: *The formula uses heights in **feet**. Both √196 = 14 and √2500 = 50 are perfect squares — ideal for an exam question. Sum = 64, then × 1.23 = 78.72. Always check: are you using feet, not metres?*

Q5. What is the minimum height for an aircraft at a range of 200 NM to be detected by a radar at 1700 ft AMSL?

a. 25,500 ft

b. 15,000 ft

c. 40,000 ft

d. 57,500 ft

✓ Correct Answer: **(b)** — 15,000 ft

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Explanation: Rearrange Range = 1.23 × (√h_{Radar} + √h_{AC}):

200 = 1.23 × (√1700 + √h_{AC})

200/1.23 = 162.6 = √1700 + √h_{AC}

√1700 ≈ 41.23

√h_{AC} = 162.6 − 41.23 = 121.37

h_{AC} = 121.37^2 ≈ 14,731 ft ≈ **15,000 ft**. See Section 4.2.

Why the other options are wrong:

- **(a)** — 25,500 ft: √25500 ≈ 159.7. Total = 159.7 + 41.23 = 200.9 → Range ≈ 247 NM. Too high — overestimates the required height.
- **(c)** — 40,000 ft: √40000 = 200. Total = 241.23 → Range ≈ 297 NM. Far too high.
- **(d)** — 57,500 ft: extremely high; corresponds to range well beyond 200 NM.

Instructor's Note: *Inverse LOS calculation. The key step is dividing 200 by 1.23 first (≈162.6), then subtracting √(radar height) before squaring. √1700 ≈ 41.2 is not a perfect square — in the exam, approximate to 2 decimal places and square the result.*

Q6. Determine which of the following statements concerning atmospheric ionization are correct:

1. The highest levels of ionization will be experienced in low latitudes
2. Ionization levels increase linearly with increasing altitude
3. The lowest levels of ionization occur about midnight
4. The E-layer is higher by night than by day because the ionization levels are lower at night

a. statements 1, 2 and 3 are correct

b. statements 1, 3 and 4 are correct

c. statements 2 and 4 are correct

d. statements 1 and 4 are correct

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✓ Correct Answer: **(d)** — statements 1 and 4 are correct

Explanation:
Statement 1 — TRUE: Ionization increases as latitude decreases (low latitudes = equatorial regions = most solar radiation intensity).
Statement 2 — FALSE: Ionization does NOT increase linearly — it forms into discrete layers due to gravitational and magnetic effects at high altitudes.
Statement 3 — FALSE: The lowest levels of ionization occur just **before sunrise**, not midnight. After midnight, ionization continues to decay but reaches minimum just before dawn.
Statement 4 — TRUE: The E-layer reduces in altitude at sunrise (more ionization from solar radiation), and increases in altitude after sunset (less ionization). So it IS higher at night. See Section 5 and Section 6.

Why the other options are wrong:

- **(a)** — Includes statement 2 (linearly — FALSE) and statement 3 (midnight — FALSE).
- **(b)** — Includes statement 3 (midnight — FALSE).
- **(c)** — Includes statement 2 (linearly — FALSE) and excludes statement 1 (TRUE).

Instructor's Note: *This compound question tests four facts simultaneously. The two most common errors: confusing "midnight" with "just before sunrise" for minimum ionization, and assuming ionization increases linearly. Both are specifically contradicted in the source text.*

Q7. The average height of the E-layer is ..... and the maximum range for sky wave will be .....

a. 60 km, 1350 NM

b. 125 km, 2200 km

c. 225 km, 2200 km

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d. 125 km, 1350 NM

✓ Correct Answer: **(d)** — 125 km, 1350 NM

Explanation: E-layer average altitude = **125 km**. Maximum E-layer sky wave range = **1350 NM**. (The F-layer at 225 km gives the 2200 NM range.) See Section 7.5.

Why the other options are wrong:

- **(a)** — 60 km is the base of the ionosphere / bottom of the D-layer, not the E-layer height. 1350 NM is correct for E-layer but paired with wrong altitude.
- **(b)** — 125 km is correct for E-layer, but 2200 km (note: km not NM!) is wrong unit AND wrong layer — 2200 NM is the F-layer range.
- **(c)** — 225 km is the F-layer height, not E-layer. 2200 km uses wrong unit (should be NM).

Instructor's Note: *Two pairs to memorise: E-layer = **125 km** → **1350 NM**; F-layer = **225 km** → **2200 NM**. Options (b) and (c) use km instead of NM for range — a unit trap. Check units in every answer option.*

Q8. Concerning HF communications, which of the following is correct?

a. The frequency required in low latitudes is less than the frequency required in high latitudes

b. At night a higher frequency is required than by day

c. The frequency required is dependent on time of day but not the season

d. The frequency required for short ranges will be less than the frequency required for long ranges

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✓ Correct Answer: **(d)** — The frequency required for short ranges will be less than the frequency required for long ranges

Explanation: For HF sky wave comms, short ranges require lower frequencies (smaller skip distances); long ranges require higher frequencies (larger skip distances needed). This is a direct and unambiguous fact. See Section 8.

Why the other options are wrong:

- **(a)** — The opposite is true. Low latitudes have higher ionization (more solar radiation), so the MUF is higher → frequency required is HIGHER in low latitudes, not lower.
- **(b)** — The opposite is true. Ionization is lower at night → lower MUF → lower frequency needed at night. A good rule of thumb: night frequency ≈ half day frequency.
- **(c)** — Completely wrong on both counts. Frequency required depends on BOTH time of day (ionization levels vary diurnally) AND season (summer ionization is higher → higher MUF).

Instructor's Note: *Options (a) and (b) are direct reversals of correct facts — classic exam traps. Option (c) contradicts both the source and basic understanding. Option (d) is directly stated in the text: "Short ranges will require lower frequencies and longer ranges will require higher frequency."*

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📊 Master Reference Tables

All Numerical Values — Chapter 2

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Value	Description	Section
$P \propto 1/R^2$	Inverse square law	§2.1
×4 power	Power needed to double range	§2.1
1000 MHz	Absorption becomes very significant above this	§2.1
20 kHz – 50 MHz	Surface wave frequency range	§4.1
30 kHz – 2 MHz	Surface wave normally used (LF–lower MF)	§4.1
Range = 1.23×(√hTX + √hRX)	LOS range formula (heights in feet, NM)	§4.2
60 km	Base of ionosphere	§5
1500 km	Notional upper limit of ionosphere	§5
0.22 hPa	Atmospheric pressure at 60 km	§5
~1400 local	Peak ionization time (daily)	§5
~75 km	D-layer average altitude	§6
~125 km	E-layer average altitude	§6
~225 km	F-layer average altitude	§6
>400 km	F2-layer max altitude in summer	§6
2 MHz	Boundary: D absorbs below this / E refracts above	§7.4
50 MHz	Upper limit of sky wave under normal conditions	§7.2
1350 NM	E-layer maximum sky wave range	§7.5
2200 NM	F-layer maximum sky wave range	§7.5
OWF = 0.85 × MUF	Optimum working frequency	§8
Night f ≈ ½ Day f	HF day/night frequency rule of thumb	§8
+40%	Super-refraction: max range increase	§10
−20%	Sub-refraction: max range decrease	§10

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Answer Key — Chapter 2 Questions

Q1	Q2	Q3	Q4
c	b	d	a
Q5	**Q6**	**Q7**	**Q8**
b	d	d	d

Mnemonics & Memory Aids

- **Inverse Square Law:** "Double range = 4× power" ($2^2 = 4$)
- **LOS formula:** 1.23 × (√TX + √RX) — heights in **feet**
- **Layers low to high:** D (75 km) → E (125 km) → F (225 km) — "DEF ascending"
- **D-layer:** Day only — "D for Day"; disappears at night → ADF unreliable at night
- **E/F ranges:** E = 1350 NM; F = 2200 NM — "E is less, F is far"
- **OWF:** 0.85 × MUF — "85% safety margin below max"
- **HF day/night:** Night ≈ half day frequency — "night needs half"
- **Super-refraction:** High pressure, warm over cool, +40%; Mediterranean/Caribbean
- **Sub-refraction:** Low pressure, cold over warm, −20%
- **Frequency ↑ → skip distance ↑, surface wave ↓ → dead space ↑↑**

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